KOR-PE-01 PE Node Failure SOP: Critical Hardware/Environmental Fault Resilience on Cisco PE Equipment

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_3681

Description

This Standard Operating Procedure (SOP) addresses the failure scenario of Node ID: KOR-PE-01, situated within the Koramangala Datacenter, operating on Cisco equipment as a Provider Edge (PE) role. The failure domain in question pertains to Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss), as signified by the critical alarm code ALARM_3681. The underlying protocol and hardware mechanics contributing to this failure domain are rooted in the PE node's reliance on ASIC-based forwarding engines, where memory parity errors can escalate into line card crashes due to the absence of adequate redundancy mechanisms within the Cisco PE equipment. Furthermore, power supply redundancy loss can exacerbate the situation, rendering the PE node incapable of maintaining a stable control plane.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Koramangala Datacenter to physically verify the failure scenario.

- Verify the status of redundant links, including any potential link aggregation or bonding configurations, to determine their operational integrity.

- Confirm the PE node's power supply redundancy status to assess whether it has fallen below the recommended minimum threshold.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support | sec ASIC, Expected Output: ASIC-related information, including error logs and diagnostic output., Description: This command isolates ASIC-related information, enabling the identification of potential memory parity errors or other hardware malfunctions.

Command: show power redundancy status, Expected Output: Power supply redundancy status, including current power levels and redundancy
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thresholds., Description: This command verifies the power supply redundancy status, helping to determine whether the PE node's power supply has fallen below the recommended minimum threshold.

Command: show linecard status, Expected Output: Line card status, including operational integrity and any error logs., Description: This command assesses the operational integrity of line cards within the PE node, helping to identify potential line card crashes or other hardware malfunctions.
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Resolution Steps

Step 1: Step: Replace Faulty ASIC, Command: process restart asic, Description: This command initiates a restart of the faulty ASIC, which should rectify memory parity errors and other hardware malfunctions.

Step 2: Step: Replace Faulty Power Supply, Command: power supply replace, Description: This command initiates the replacement of the faulty power supply, which should restore power supply redundancy and prevent future failures.

Step 3: Step: Replace Faulty Line Card, Command: linecard replace, Description: This command initiates the replacement of the faulty line card, which should rectify line card crashes and other hardware malfunctions.

Post-Check Verification

```
Command: show tech-support | sec ASIC, Expected Output: No ASIC-related errors or malfunctions., Description: This command verifies that the faulty ASIC has been successfully replaced and operational integrity has been restored.

Command: show power redundancy status, Expected Output: Power supply redundancy status indicates a healthy and redundant power supply configuration., Description: This command verifies that power supply redundancy has been restored and the PE node is operating within recommended minimum thresholds.

Command: show linecard status, Expected Output: Line card status indicates operational integrity and no errors., Description: This command verifies that the faulty line card has been successfully replaced and operational integrity has been restored.
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Roll-back Procedures

- Step: Restore Original Configuration, Command: copy running-config startup-config, Description: This command restores the original configuration, reversing any changes made during the resolution process.

- Step: Revert to Previous Power Supply Configuration, Command: power supply revert, Description: This command reverts to the previous power supply configuration, restoring the original power supply redundancy status.

- Step: Revert to Previous Line Card Configuration, Command: linecard revert, Description: This command reverts to the previous line card configuration, restoring the original line card status.